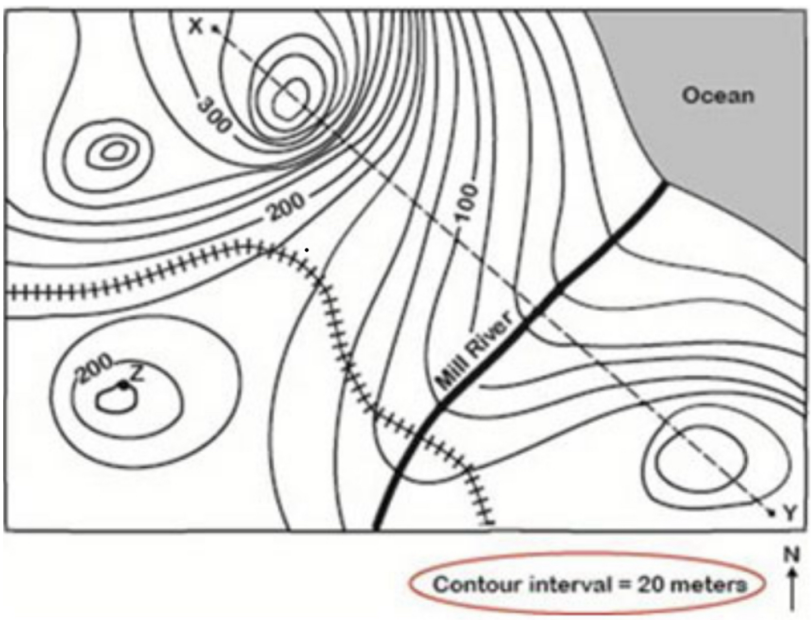
Worksheet 6 - Gradient, Taylor series, the 2nd derivative test.

1. Four sharks are at the origin of coordinates when they smell blood in the water. The first shark swims 10 km/hr at an angle 37° east of true north and finds the concentration of blood is increasing at a rate of 20 (ml/m^3)/min. The second shark swims 15 km/hr at an angle 53° west of true north and finds the concentration is decreasing at a rate of 15 (ml/m^3)/min.

   (a) The third shark swims southeast at 20 km/hr. At what rate does it find the concentration of blood is changing?

   (b) In which direction should the fourth shark swim so as to make the concentration increase as quickly as possible?

2. Consider the contour map shown below. Based on what you know about gradients (and rivers), what is the angle between the river and the contours? Why must this be the case?



3. Consider $f(x,y) = 2\cos x - y^2 + e^{xy}$.

   (a) Show that $(0,0)$ is a critical point for f.

   (b) Calculate each of f_{xx}, f_{xy}, f_{yy} at $(0,0)$ and use this to write out the 2nd-order Taylor approximation for f at $(0,0)$.

   (c) To make sure the next two problems go smoothly, check your answer to (b) with the instructor.

4. Let $g(x,y)$ be the approximation you obtained for $f(x,y)$ near $(0,0)$ in 1(b). It's not clear from the formula whether g, and hence f, has a min, max, or a saddle at $(0,0)$. Test along several lines until you are convinced you've determined which type it is. In the next problem, you'll confirm your answer in two ways.

5. Consider alternate coordinates (u,v) on $\mathbb{R}^2$ given by $(x,y) = (u-v, u+v)$.

(a) Sketch the u- and v-axes relative to the usual x- and y-axes, and draw the points whose (u, v)-coordinates are: $(-1, 2)$, $(1, 1)$, $(1, -1)$.

(b) Express g as a function of u and v, and expand and simplify the resulting expression.

(c) Explain why your answer in 3(b) confirms your answer in 2.

(d) Sketch a few level sets for g. What do the level sets of f look like near $(0, 0)$?

(e) It turns out that there is always a similar change of coordinates so that the Taylor series of a function f which has a critical point at $(0, 0)$ looks like $f(u, v) \approx f(0, 0) + au^2 + bv^2$. In fact this is why the 2nd derivative test works.

Double check your answer in 2 by applying the 2nd-derivative test directly to f.